

5. (a) A law of motion can be expressed as:

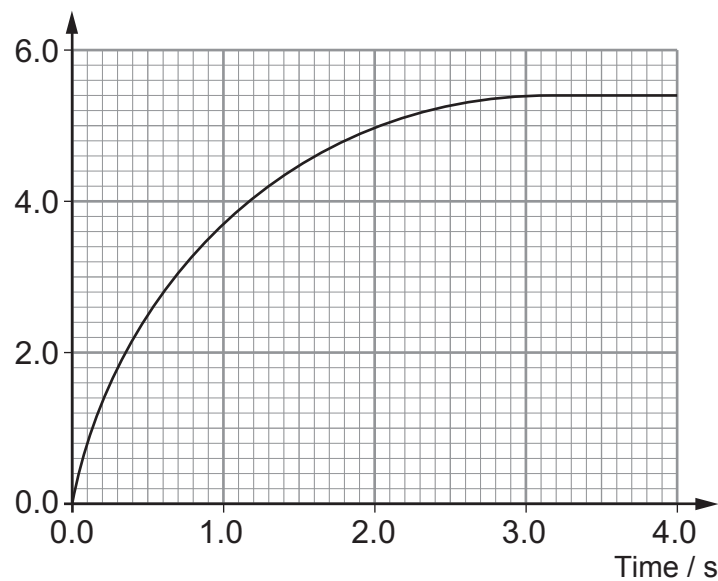
$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

State the name of the law.

[1]

- (b) The graph shows how the momentum of a spacecraft varies with time.

Momentum / 10^3 kg m s^{-1}



- (i) By drawing a suitable tangent, show that the resultant force on the spacecraft at $t = 1.0 \text{ s}$ is approximately 2 kN . [2]

- (ii) Hence show that the mass of the spacecraft is approximately 5000 kg , given that its acceleration at $t = 1.0 \text{ s}$ is 0.4 m s^{-2} . [1]

- (iii) **Label, with the letter P,** a point on the graph where the resultant force on the spacecraft is zero. [1]



- (c) At $t = 4.0\text{ s}$ the spacecraft 'docks' (collides) with another **stationary** spacecraft of mass $7\,000\text{ kg}$. They join on impact.

(i) State the principle of conservation of momentum. [2]

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(ii) Calculate the velocity of both spacecraft after colliding. [3]

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